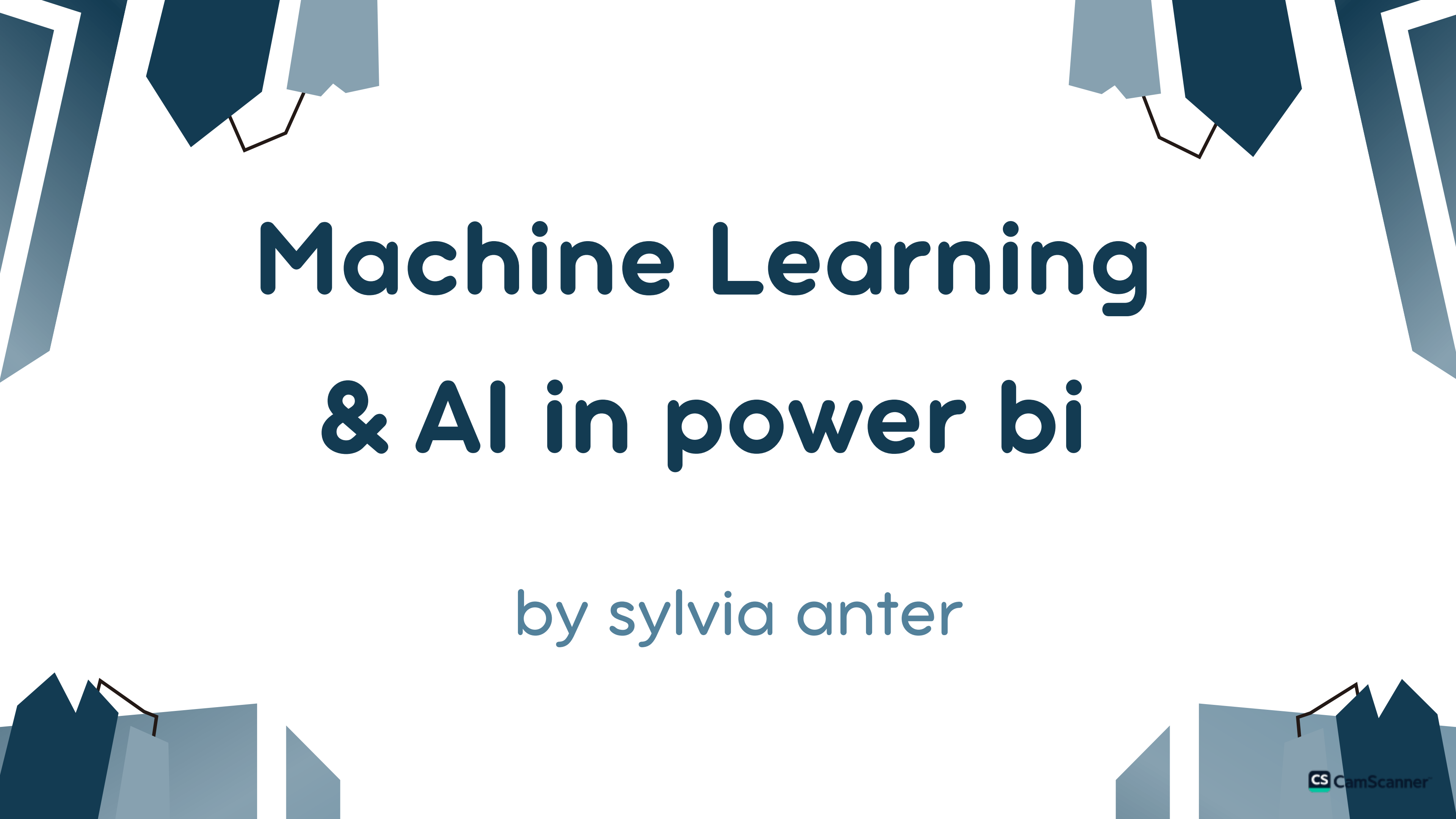




Machine Learning & AI in power bi

by sylvia anter

Why link between power bi and ML ?

- big companies don't depend on reports only
- it also depend on predictions , Explanations , Automation

Example :

power bi –

sales increased by 20 % this month

power bi + ML –

- “Sales are expected to increase by 15% next month”
- “This is mainly driven by product ‘X’”
- “We should focus on this group of customers”

Why link between power bi and ML ?

If ML do all that why power bi ?

ML Model inside notebook , VS code

this means :

Require a technical person to excute it .

it is difficult for business stakeholders to understand .

When linking between power bi and ML

managers , stakholders don't open python

but can open dashboard they can see :

1- predictions

2- visuializations

3- KPIS

Why link between power bi and ML ?

Machine Learning = Brain (predictions)

Power bi = eyes (decision & visulization)

ML – Technical work only

ML + Power bi used by :

business teams , marketing team , sales team ...

COURSE CONTENT

- I intro to power bi
- 2 Ai Graphics
- 3 Install python with power bi
- 4 Advanced Analytics using python
- 5 Machine Learning Revision
- 6 Deploy Models in power bi
- 7 Scikit-Learn
- 8 AutoML with python & Pycaret
- 9 Regression Models with python in power bi
- IO Classification models with python in power bi
- II clustering models with python in power bi

Intro To power bi

Power BI Desktop Interface



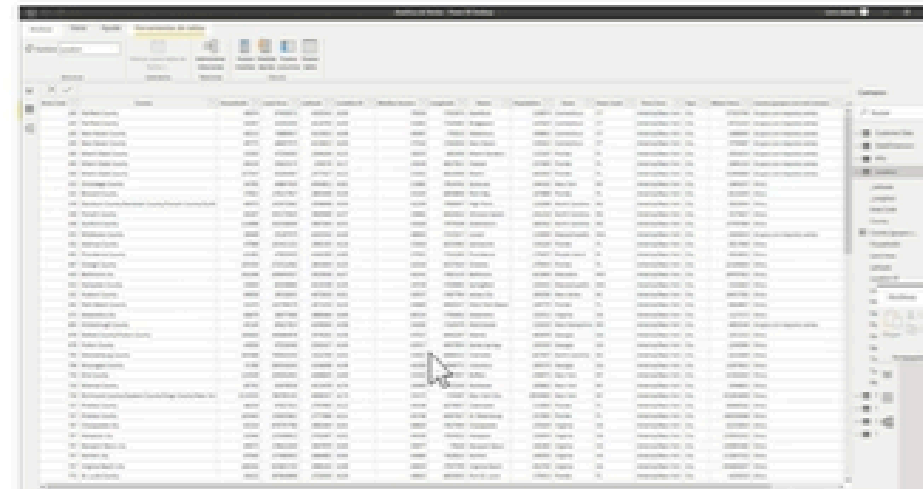
Main Windows:

- Report 
- Data 
- Model 

- power bi is a business intelligence visualization tool from microsoft , desktop & service (cloud application)

- 1- Report : Dashboard
- 2- Data : Tables
- 3- Model : Relationship between Tables

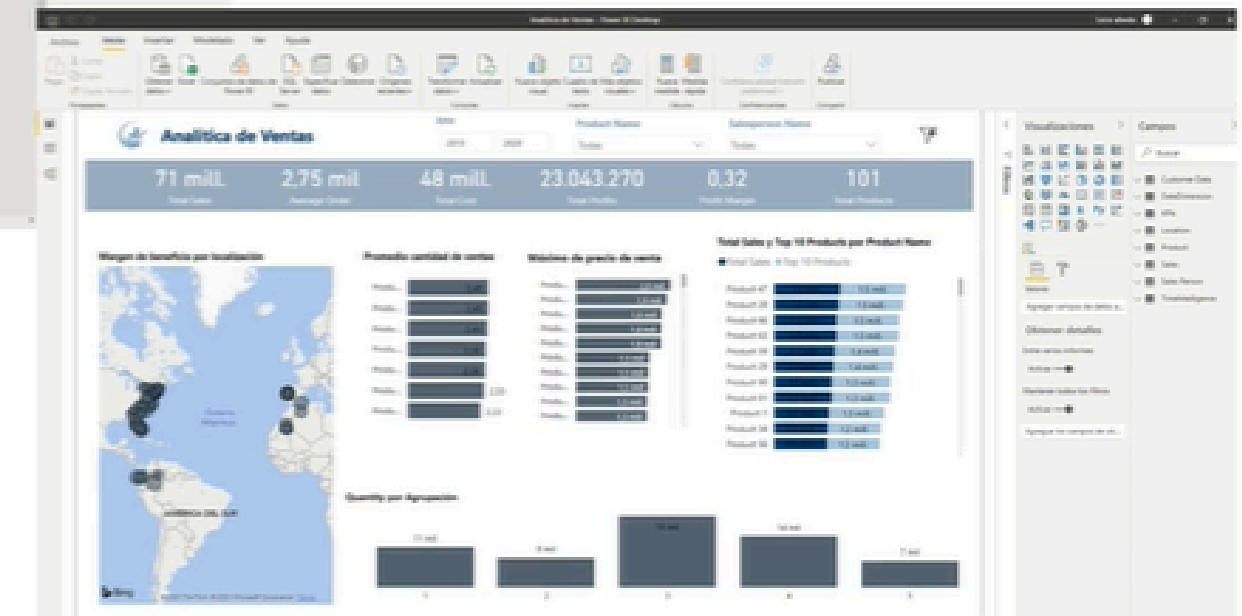
Workflow in Power BI



Connecting, modeling and transforming data

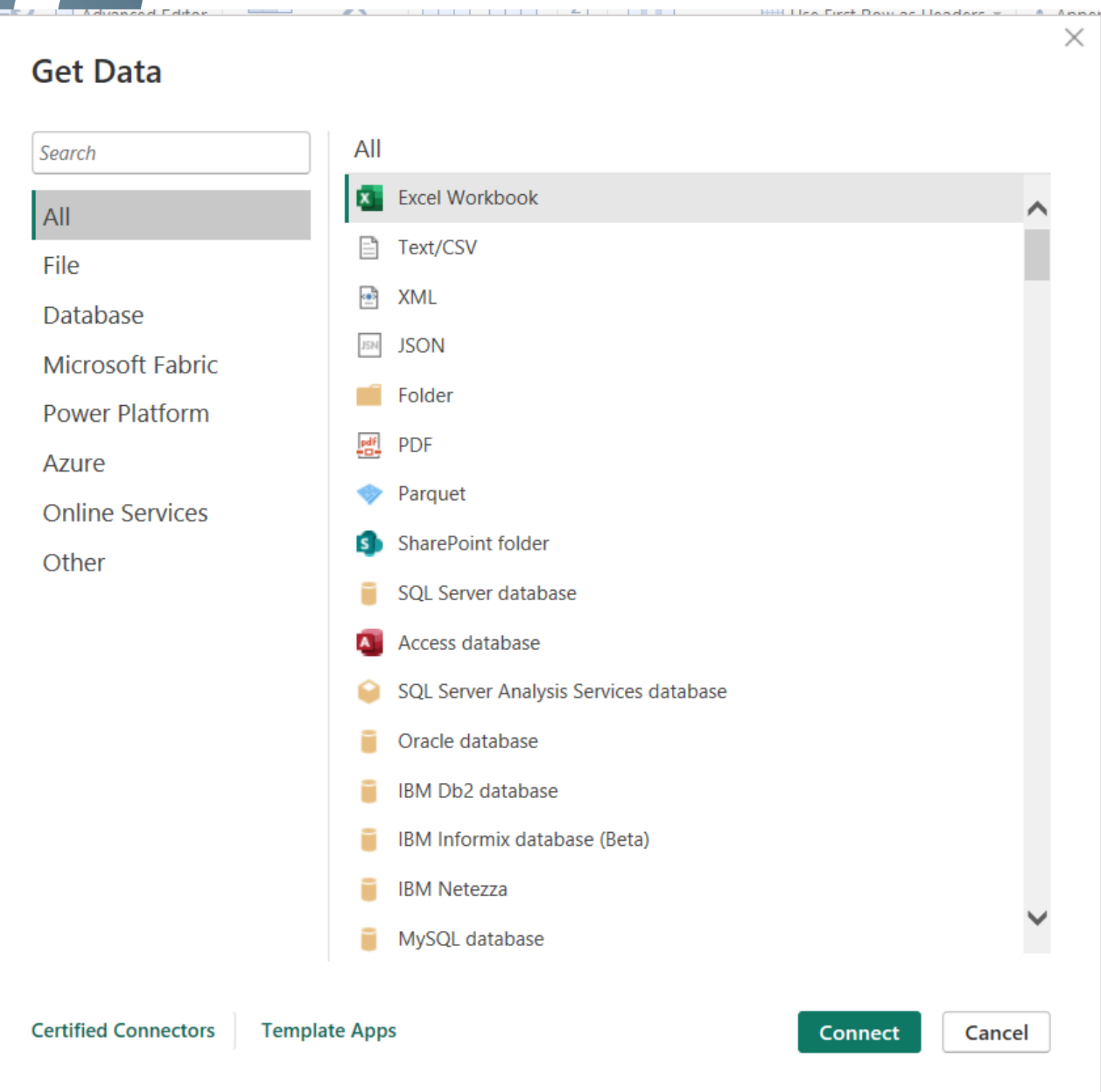


Modeling: creating relationships between tables



Designing the report to visualize and analyze the data

IMPORT DATA



- From Get Data : we Can Get data From Excel , CSV , TXT , PDF , Json , XML , Database , Python Script

IMPORT DATA

- after we upload the files , choose close and apply to load the data

The screenshot displays the Microsoft Power Query Editor interface. The ribbon at the top includes tabs for File, Home, Transform, Add Column, View, Tools, and Help. The Transform tab is active, showing various data manipulation options like Close & Apply, New Source, Recent Sources, Enter Data, Data source settings, Manage Parameters, Export query results, Refresh Preview, Properties, Advanced Editor, Manage, Choose Columns, Remove Columns, Keep Rows, Remove Rows, Sort, Split Column, Group By, Data Type: Whole Number, Use First Row as Headers, Replace Values, Merge Queries, Append Queries, and Combine Files.

On the left, the 'Queries [5]' pane lists 'Channel', 'District', 'Item', 'Sales', and 'Store'. The main workspace shows a data preview for the 'Channel' query. The formula bar at the top displays the M code: `= Table.TransformColumnTypes("#Promoted Headers",{{"ChannelID", Int64.Type}, {"ChannelName", type text}})`.

The data preview table is as follows:

	ChannelID	ChannelName
	Valid 100%	Valid 100%
	Error 0%	Error 0%
	Empty 0%	Empty 0%
1	1	Store
2	2	Online
3	3	Catalog
4	4	Reseller

On the right, the 'Query Settings' pane shows the 'Channel' query name and a list of 'APPLIED STEPS': Source, Promoted Headers, and Changed Type.